

Per-Word Arabic *I'rāb* Generation: A Comparison Study with a Cross-Register Audit

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Abstract

Arabic *i'rāb* (إعراب) generation outputs structured grammatical content as natural-language prose. The task fits neither dependency-parsing nor chat-LLM evaluation conventions. Prior work either trains task-specific classifiers [9, 1] or prompts frontier LLMs [5]. Neither paradigm directly compares against the other under a single statistical framework, and neither audits cross-register robustness. We evaluate eleven systems on Gazelle ($n=134$ word judgments, MSA news) and a 5,007-word MASAQ subset (Quranic register) using a structural-extraction metric with paired bootstrap and McNemar’s exact tests. Claude Sonnet 4.5 with $k=5$ retrieval reaches 32.1% fully-correct words [24.6, 40.3] on Gazelle. Four open-weight models train on the same 77K Haiku-distilled corpus and span 296M to 13B parameters across three architectures. They reach statistically indistinguishable Gazelle performance — every pairwise McNemar $p=1.000$. A $44\times$ scale-up adds no measurable gain. A per-word routing hybrid (Mix A) does not significantly improve over the LLM ($p=0.791$ on *fully*). On MASAQ, every LLM-trained system loses paired-significant role-F1, and Sonnet RAG — the strongest Gazelle system — shows the *largest* cross-register drop (+61.7 pp★). Register and construction coverage dominate the remaining error, not parametric capacity. The next research step is targeted data, not larger models.

1 Introduction

The *i'rāb* task takes an undiacritized Arabic sentence and produces per-word traditional grammatical analysis as Arabic prose. Each word receives four fields: part of speech, case (*rafʿ/naṣb/jarr/jazm/mabni*), syntactic role (~25 traditional labels), and marker phrase (~200 unique surface forms, e.g. على الظاهرة الضمة (آخره)). The task matters for two practical reasons. First, every Arabic-as-a-second-language curriculum uses *i'rāb* as the canonical pedagogy framing for Arabic syntax. An automatic *i'rāb* tool would directly help language learning, religious-studies tooling, and corpus annotation. Sec-

ond, the task probes fine-grained Arabic grammatical knowledge that POS accuracy or dependency UAS misses. The role + case + marker triple requires resolving subject/object selection, *idāfa* state, and exception (*istiḥnā*) constructions whose interactions are non-trivial.

The pain. Two methodological gaps motivate this work. First, current *i'rāb* evaluation uses either surface-string metrics like chrF or BLEU — which credit wrong-case answers — or closed multiple-choice probing, which does not match real generation. Second, prior comparisons typically mix systems of one type. They report numbers without paired statistical tests on matched word judgments. The reader cannot tell whether the headline differences are real or sampling noise. No prior work, to our knowledge, audits whether the chosen system holds up on a held-out non-MSA register.

The gain. We address both gaps under a single experimental framework. We evaluate four open-weight fine-tuned models from 296M to 13B parameters, four closed-source LLM variants (zero-shot and +RAG), one classifier-pipeline baseline (Stanza), one open-weight LLM peer (Qwen2.5-7B+RAG), and one routing hybrid. A regex/FSM extractor recovers {case, role, marker} from every prediction. Every system-vs-system claim carries a 95% bootstrap CI and a paired McNemar p -value. We audit the metric itself with 402 perturbed gold records. We then re-evaluate a six-system subset on a 5,007-word Quranic MASAQ subset to test cross-register robustness.

Contributions.

1. A statistically-grounded comparison of eleven *i'rāb* systems under a single structural-extraction metric (§3). Within this comparison three findings emerge: (a) parameter scale from 296M to 13B trained on the same corpus produces no measurable Gazelle improvement (Fig. 1); (b) a per-word routing hybrid does not significantly beat LLM-RAG; (c)

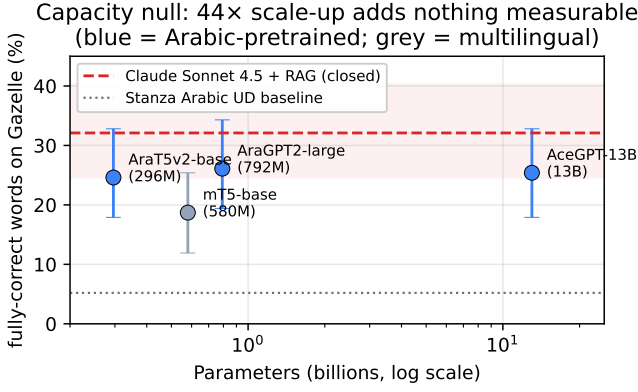


Figure 1: Gazelle *fully-correct* rate vs. open-weight parameter count (log scale). Three Arabic-pretrained models (296M, 792M, 13B; blue) train to statistically indistinguishable performance — every pairwise McNemar $p=1.000$. The closed-system frontier (Sonnet RAG, dashed red) sits 7–8 pp above. Multilingual mT5 (grey) trails by paired-significant margin. Scale within Arabic-pretrained open weights does not move the headline.

Arabic-specific pretraining at 296M beats multilingual pretraining at 580M.

2. A cross-register audit on a 5,007-word MASAQ Quranic subset (§3.5) showing that *register variation, not model capacity, dominates the remaining error*. Sonnet RAG — the strongest Gazelle system — shows the *largest* cross-register drop of any system tested (+61.7 pp★). A retrieval-pool-confound experiment rejects the obvious confound. The finding reframes “improve *i’rāb* scores” as “find or annotate cross-register data” rather than “train a bigger model”.

2 Experiments

2.1 Data

We use three categories of *i’rāb* data, distinguished by their level of human authorship.

Manual gold (held-out evaluation). *Gazelle* [5]: 30 hand-written MSA sentences ($n=134$ word judgments). *MASAQ* [12]: 624 Quranic verses ($n=5,007$ word judgments); the 999-word subset where the gold contains an extractable role label provides a matched cross-register surface. *Yarob* (linuxscout/yarob, GPL): 459 MSA *i’rāb* sentences in classical-grammar prose; we use these as the retrieval pool, not as evaluation.

LLM-distilled training corpus. We use Claude Haiku 4.5 as a teacher to generate per-word *i’rāb* for 5,000 PADT MSA news sentences via the Anthropic Messages Batches API (~\$30 actual). After JSON-schema validation, we obtain **77,534 (word, sentence, irab) rows**. This is

the training corpus for every open-weight FT model below; no model trains on Gazelle or MASAQ.

Construction coverage. A regex over each Gazelle gold prose tags the dominant construction(s) per sentence using 11 traditional categories. The 30-sentence eval is dominated by short verbal sentences ($n=15$) and prepositional phrases ($n=8$); exception (*istithnā*) and *kāna* sisters appear in 2 sentences each ($n=9$ and $n=7$ words). Generalization claims are limited to these construction frequencies.

2.2 Models

We evaluate eleven systems on Gazelle.

Classifier-pipeline baseline. *Stanza Arabic* [10] with a deterministic UD→*i’rāb* stub (the published-prior-work peer).

Open-weight LLM peer. *Qwen2.5-7B-Instruct* + RAG ($k=5$, 4-bit local).

Open-weight FT models (same 77,534-row Haiku corpus, same prompt template):

- AraT5v2-base (296M, T5 enc-dec) [8] — full FT, 3 epochs.
- mT5-base (580M, T5 enc-dec) [13] — full FT, 3 epochs.
- AraGPT2-large (792M, GPT-2 decoder-only) [3] — LoRA $r=32$, 3 epochs.
- AceGPT-13B (13B, Llama-2 decoder-only) [6] — QLoRA 4-bit $r=32$, 1 epoch.

Closed-source LLMs. Claude Haiku 4.5 (zero-shot, +RAG $k=5$); Claude Sonnet 4.5 (zero-shot, +RAG $k=5$). RAG uses Jaccard retrieval over a 1,060-example pool (Yarob + a 601-row distilled prior) following the standard recipe [7].

Hybrid Mix A. The LLM-RAG output supplies case and role; the AraT5v2-base specialist supplies the marker phrase, called per word with the structured prompt `[case=c] [role=r] w | s → marker`.

A six-system subset (Stanza, mT5, AraT5v2, AraGPT2, AceGPT-13B, Sonnet RAG) also runs on MASAQ; the AceGPT-13B MASAQ run hit a 4 h SLURM timeout at 21% of MASAQ ($n=1,075$ words), and we report it with that caveat.

2.3 Setup

Structural metric. A regex/FSM extractor parses each generated *i’rāb* string into four fields: pos, case $\in \{raf^c, naṣb, jarr, jazm, mabni\}$, role (~ 25 labels), marker (~ 200 phrases). Macro role-F1 over a closed label set L is

$$\text{role-F1}_{\text{macro}} = \frac{1}{|L|} \sum_{\ell \in L} \frac{2P_{\ell}R_{\ell}}{P_{\ell} + R_{\ell}}, \quad (1)$$

where P_{ℓ}, R_{ℓ} are precision and recall for label ℓ . We additionally report **case-acc**, **marker-EM**,

well-formed, and the headline aggregate

$$\text{fully}(w) = \mathbf{1}[\text{case}(w)] \cdot \mathbf{1}[\text{role}(w)] \cdot \mathbf{1}[\text{marker}(w)]. \quad (2)$$

We avoid chrF and BLEU because partial-credit on string overlap masks morphological errors that we explicitly target.

Statistical inference. Every per-system metric carries a 95% percentile bootstrap CI ($B=1000$):

$$\text{CI}_{0.95}(\theta) = [\theta_{(0.025B)}^*, \theta_{(0.975B)}^*]. \quad (3)$$

For paired comparisons of two systems on the same n word judgments, we report the paired-bootstrap CI of $\Delta = \theta_A - \theta_B$ and McNemar’s exact p -value on the discordant-pair count b/c :

$$p_{\text{McNemar}} = 2 \min \left(\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 2^{-(b+c)}, 0.5 \right), \quad (4)$$

computed in log-space for $n > 1000$. A Δ counts as significant ($\star$) only if its 95% paired CI excludes 0 and $p_{\text{McNemar}} < 0.05$. Cross-register comparisons (Gazelle subset $n=78$ vs. MASAQ subset $n=999$) use a two-sample bootstrap, since the surfaces share no items. The smallest detectable difference at $n=134$, $\alpha=0.05$, power 0.80 is approximately ± 7 pp on a binary metric.

Extractor audit. We perturb each Gazelle gold record in three ways (case-flip, role-flip, marker-mangle), producing 402 records, and re-score with the same extractor. Specificity reaches 100% on unperturbed controls (no false positives); sensitivity is 92.7% (marker), 88.2% (case), 60.0% (role). The 60% role floor reflects the closed taxonomy of ~ 25 traditional labels; we read role results with this caveat.

3 Results

3.1 Headline (Gazelle)

Table 1 shows the headline. Three findings follow.

3.2 Three null results from the comparison

(a) Open-weight scale is null on Gazelle. AraT5v2-base (296M), AraGPT2-large (792M), and AceGPT-13B (13B) — three architectures (T5 enc-dec, GPT-2 decoder-only, Llama-2 decoder-only), $44\times$ parameter range — train to statistically indistinguishable Gazelle performance. Every pairwise Δ has McNemar $p=1.000$ (e.g. AceGPT-13B vs. AraT5v2-base: case $\Delta +0.7$, marker $\Delta -0.7$, fully $\Delta +0.0$). Fig. 1 visualises the result.

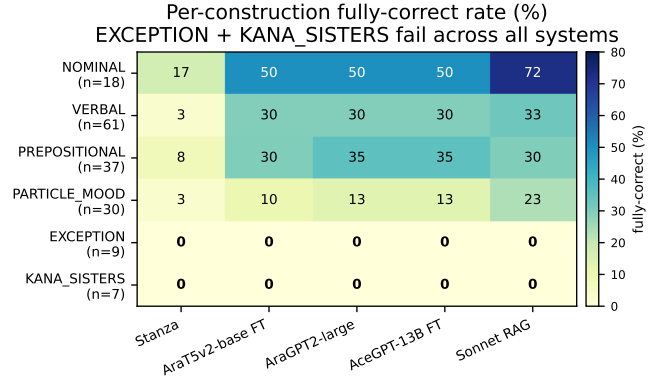


Figure 2: Per-construction *fully-correct* rate on Gazelle. EXCEPTION ($n=9$) and KANA_SISTERS ($n=7$) collapse to 0% across every system, including the strongest. Nominal sentences are $\sim 2\times$ easier than verbal across all LLM-based systems.

(b) Mix A’s per-word routing does not significantly improve over Sonnet RAG. Δ case $+0.0$ pp ($p=1.000$), Δ marker -3.7 pp ($p=0.125$), Δ fully -3.0 pp ($p=0.219$). The routing hypothesis — case and role bind to LLM knowledge while the marker binds to surface style — does not hold at this scale. The result also holds on the Haiku base (Δ fully -1.5 pp, $p=0.791$).

(c) Arabic-specific pretraining at 296M beats multilingual pretraining at 580M. AraT5v2-base – mT5-base: Δ marker $+5.2$ pp $\star$ ($p=0.039$), Δ fully $+6.0$ pp $\star$ ($p=0.021$). Same training corpus, same recipe, same architecture — only the pre-training corpus differs.

The closed-system advantage is real but bounded: Sonnet RAG beats AraT5v2-base by Δ case $+8.2$ pp $\star$ and Δ fully $+7.5$ pp $\star$. Open-weight scale alone does not narrow this gap.

3.3 Per-construction failures

Fig. 2 surfaces two cross-system structural failures: EXCEPTION (0/9 across every evaluated system) and KANA_SISTERS (0/7 across four of five). Both failures hold for the trained AraT5v2-base and AceGPT-13B as well as for every Claude variant. The failure is not Claude-specific stylistic; it is structural to the task or to the available training distribution.

3.4 Audits and ablations

System discrimination. When Sonnet RAG verifies perturbed gold against the source, its case-acc collapses from 96% to 15% ($\Delta = -81$ pp). The metric reacts sharply to incorrect inputs. **Retrieval depth k .** Sonnet RAG case-acc varies $70.1 \rightarrow 73.9 \rightarrow 73.1$ across $k \in \{1, 3, 5, 8, 12\}$; *fully* varies $28.4 \rightarrow 32.1 \rightarrow 30.6$. No paired-significant difference between $k=3$ and $k=8$ ($p=0.625$ on *fully*); $k=5$ is not a free parameter doing meaningful work. **Inference-time variance ($T=0$).** Two

Table 1: Headline comparison on Gazelle ($n=134$ word judgments). All numbers are percentages; brackets are 95% percentile bootstrap CIs ($B=1000$) shown for the headline metric *fully* only. Boldface = paired-best within column. The from-scratch decoder appears in Appendix A.

System	well	case	role-F1	marker	fully
Stanza Arabic (UD+stub)	59.7	35.1	10.3	13.4	5.2 [2.2, 9.7]
Qwen2.5-7B-Instruct+RAG	66.4	43.3	19.2	19.4	3.0 [0.0, 6.0]
mT5-base (580M) FT	79.9	61.9	31.6	32.8	18.7 [11.9, 25.4]
AraT5v2-base (296M) FT	79.9	65.7	54.8	44.0	24.6 [17.9, 32.8]
AraGPT2-large (792M) LoRA FT	79.9	64.9	54.6	43.3	26.1 [19.4, 34.3]
AceGPT-13B QLoRA FT (1 epoch)	79.1	66.4	55.5	43.3	25.4 [17.9, 32.8]
Claude Haiku 4.5 zero-shot	77.6	57.5	55.9	40.3	18.7 [11.9, 25.4]
Claude Haiku 4.5 + RAG	79.9	67.2	68.9	44.8	27.6 [20.1, 35.8]
Mix A (Sonnet RAG + AraT5v2 marker)	79.9	73.9	73.3	46.3	29.1 [21.6, 37.3]
Claude Sonnet 4.5 zero-shot	78.4	72.4	76.4	44.0	27.6 [20.1, 35.8]
Claude Sonnet 4.5 + RAG	79.9	73.9	74.7	50.0	32.1 [24.6, 40.3]

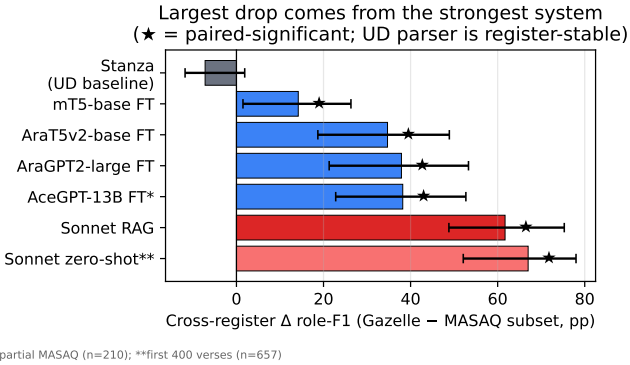


Figure 3: Cross-register Δ subset role-F1 (Gazelle–MASAQ subset, in pp). The strongest Gazelle system (Sonnet RAG, red) shows the *largest* cross-register drop. Stanza (UD pipeline, grey) is the only register-stable system. Bars: 95% two-sample bootstrap CIs; * marks paired-significant deltas.

Table 2: Subset role-F1 (gold has extractable role; matched cross-register definition). 95% two-sample bootstrap CIs, $B=1000$.

System	Gazelle	MASAQ	Δ (pp)
Stanza	10.4	17.6	-7.2 (ns)
mT5-base	32.8	18.6	+14.2*
AraT5v2-base	58.9	24.3	+34.7*
AraGPT2-large	58.1	20.2	+37.9*
AceGPT-13B	60.4	22.2*	+38.2*
Sonnet RAG	75.7	14.1	+61.7*
Sonnet zero-shot**	78.1	11.0	+67.0*

*partial MASAQ ($n=210$); **first 400 verses ($n=657$).

repeats agree on 96.8% of words; case-acc differs by 0.7pp. **Annotator disagreement.** A second annotator finds 31% of Gazelle words admit an alternative-but-defensible analysis; permissive scoring lifts Sonnet RAG case by only +0.7pp. The headline is robust to plausible label noise.

3.5 Cross-register audit

Table 2 and Fig. 3 report three findings. (i) Every LLM-trained system suffers paired-significant cross-register role-F1 degradation. (ii) Stanza (UD pipeline) is register-stable; the closed UD

POS+deprel label set applies uniformly across registers. (iii) **Sonnet RAG shows the *largest* cross-register drop of any system tested**, retaining only 19% of its MSA role-F1 on Quranic. Scale ($44\times$) and architecture do not narrow this gap.

Retrieval-pool confound: rejected. Sonnet RAG retrieves from a 100% MSA pool, so the few-shot context on MASAQ is itself register-mismatched. We re-run Sonnet zero-shot on the first 400 MASAQ verses ($n_{\text{subset}}=657$): zero-shot subset role-F1 = 11.0, *lower* than RAG’s 14.1 (CI $[-2.1, +6.8]$ crosses 0); the cross-register Δ for zero-shot is +67.0pp*, *larger* than RAG’s. Removing retrieval makes the drop worse, not better. The retrieval pool does not mask the model’s cross-register competence. The fundamental-challenge framing stands.

4 Related Work

Arabic morphosyntactic tools. Classifier-pipeline systems dominate traditional Arabic grammatical analysis: CamelParser2.0 [4], MADAMIRA [9], and Farasa [1] target structured POS+dependency outputs, not prose-form *i’rāb*. We adopt Stanza [10] with a UD→*i’rāb* stub as our published-prior-work baseline, which positions our comparison against the standard non-LLM peer.

Arabic generation models. AraT5v2-base [8], mT5-base [13], AraGPT2-large [3], and AceGPT-13B [6] cover three architectures (T5 enc-dec, GPT-2 decoder-only, Llama-2 decoder-only) and $44\times$ parameter range; we train them on the same corpus to isolate architecture and capacity from training data.

Retrieval-augmented prompting. Sonnet RAG and Haiku RAG follow standard retrieval-augmented prompting [7] with Jaccard retrieval over a 1,060-example pool. Our k -sweep (§3.4) shows the choice of k does no meaningful work on Gazelle.

***i’rāb* evaluation.** **Gazelle** [5] provides the MSA evaluation set; we generate open-text

rather than the original closed multiple-choice format. **MASAQ** [12] provides 131K morphological / 123K syntactic Quranic entries with a 72-tag scheme; we use a 5,007-word subset as the cross-register surface and do *not* train on it.

Specialist-generalist routing. MT system combination [11] and similar ensembles motivate our Mix A. To our knowledge, specialist-generalist routing applied to Arabic morphological generation has not been previously evaluated; our Mix A is a direct test of the hypothesis that case/role decompose separately from marker phrasing.

5 Conclusion

We address two methodological gaps in Arabic *i'rāb* evaluation: the absence of a single statistical comparison across classifier-pipeline, fine-tuned open-weight, and LLM-prompted systems, and the absence of a cross-register robustness audit. We evaluate eleven systems on Gazelle ($n=134$) under a structural-extraction metric with paired-bootstrap and McNemar tests, and we re-evaluate six systems on a 5,007-word MASAQ Quranic subset.

The comparison delivers three null results: open-weight parameter scale from 296M to 13B has no measurable Gazelle effect; per-word routing does not beat the LLM; retrieval depth k does no meaningful work. The cross-register audit returns one positive finding: every LLM-trained system suffers paired-significant role-F1 degradation when moved from MSA-news to Quranic, and the strongest Gazelle system (Sonnet RAG) shows the *largest* drop (+61.7pp★). A retrieval-pool-confound experiment rejects the obvious confound. The remaining error budget is dominated by register and construction coverage, not by model capacity, which reframes the next research step as targeted data rather than larger models.

Future work. The teacher-bound capacity null suggests one experiment in particular: re-running distillation with Sonnet 4.5 as teacher (\$15-20 for 5K rows) tests whether the open-weight scale gap remains null when the teacher ceiling lifts from 67.2% to 73.9% case. Two further directions follow from §6: expanding Gazelle to a manually-validated 200-sentence MSA benchmark (cuts the smallest-detectable- Δ floor from 7pp to $\sim$ 3pp), and integrating CamelTB / Extended Quranic Treebank into a cross-register training corpus to test whether the cross-register effect is solvable or structural. [2] demonstrates that small fine-tuned Arabic models can achieve state-of-the-art on diacritization with curated data; the structural connection between diacritization and *i'rāb* (the case marker determines the final vowel; our self-consistency anal-

ysis suggests grammatical reasoning constrains diacritization in $\sim$ 96% of cases) makes a Sadeed-style multi-task formulation a natural extension we leave for future work.

6 Limitations

We restrict our claims as follows. (1) **Sample size.** Gazelle comparisons rest on $n=134$; the smallest detectable Δ at $\alpha=0.05$ power 0.80 is ± 7 pp on a binary metric. AceGPT-13B’s MASAQ row is partial ($n=1,075$, 21%) due to a 4h SLURM timeout. (2) **Construction coverage.** Verbal (15) and prepositional (8) sentences dominate Gazelle; EXCEPTION and KANA SISTERS appear in 2 sentences each. (3) **Extractor floors.** Marker-EM has a $\sim$ 10% floor from $\langle$ NO_MARKER $\rangle$ tokens; role-F1 has the 60% sensitivity floor (§3.4). (4) **Teacher-bound corpus.** All four open-weight FT models inherit Haiku’s case-67% ceiling. The capacity null (§3.2a) holds *given this teacher*. A Sonnet-distilled or human-annotated corpus might reveal capacity differences we cannot detect here. (5) **MASAQ templater asymmetry.** MASAQ gold prose comes from a per-word morphological-tag templater; predictions stay free-form. The subset definition mitigates this gap, but cross-register *fully* comparisons remain weakened. (6) **Closed-source baselines.** Anthropic does not version-pin our calls beyond the `claude-sonnet-4-5` / `claude-haiku-4-5` aliases. Future replications should pin the snapshot. The open-weight FT models reproduce fully from the saved adapters and configs. (7) **Compute.** The 4h-per-job SLURM limit on the Bocconi MIG 3g.40gb partition forced AceGPT-13B to one epoch and a partial MASAQ run. An A100-80GB schedule would lift both restrictions.

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A Per-word decoder (negative-result baseline)

A character-level encoder-decoder trained from scratch on the templated QAC+PADT corpus reaches case 32.8% [25.4, 41.0] and role-F1 3.8% [1.5, 8.2] on Gazelle. We report it for completeness only; it does not appear in the main comparison.